

Client context: Healthcare revenue cycle management (RCM) platform serving 300+ behavioral health facilities.

- **Claims Reconciliation & Discrepancy Analysis:** Built a reconciliation pipeline across CollaborateMD and internal RCM ledgers using normalized 12-field exact compound keys, Institutional vs. Professional claim-type rules, and bucketed similarity-based candidate matching; generated discrepancy sheets linking directly to source records for operator review, shrinking facility reconciliation and migration timelines from over a month to three days.
- **Dynamic Query Compiler & Hybrid AI Workflows:** Built a recursive query compiler in TypeScript that translates MongoDB-style JSON filter trees (\$and, \$or, \$in, \$like) into parameterized MySQL predicates against an allowlisted column schema; used Gemini in Google Sheets to convert unstructured plain-English audit requests into validated JSON expressions, enabling non-technical compliance auditors to execute dynamic medical-record audits without writing SQL.
- **Productivity Analytics & Product Prototyping:** Built a workforce productivity engine computing 15 operational metrics across 8 targeted SQL queries per specialist, including claim touches, co-toucher splits, and payment attribution; reduced JDBC result-access overhead using server-side JSON aggregation (JSON_ARRAYAGG) on a MySQL replica to retrieve one structured result per query for client-side processing. The prototype was subsequently productized into Zion’s enterprise web application.
- **Prefix Discovery & Transactional Migrations:** Built an automated insurance prefix discovery pipeline that mined prefix-to-payer mappings directly from the Blue Cross Blue Shield clearinghouse website, distributing 46,656 candidate lookups across seven daily buckets using a deterministic modulo-7 hash; built a five-stage migration workflow (Discover, Snapshot, Transform, Prepare, Run) with per-patient SQL transactions, generated-ID propagation, and reverse-order compensating undo statements.
- **Serverless Privilege Gateway & Reporting Infrastructure:** Built an intermediary gateway service enabling non-technical operations staff to run database reports from Google Sheets without direct database credentials; routed requests through an execution boundary that translated unprivileged user calls into authorized service-identity queries, using asynchronous trigger queues to decouple long-running SQL executions and write formatted datasets back to destination sheets.
- **Developer Tooling & AI Workflows:** Built a Python CLI using uv, Click, and Pydantic managing database access across dev, QA, and production environments, with environment-specific TLS client certificates, OS keyring credential storage, and read-only session defaults; added an interactive terminal (TTY) interlock that rejects automated non-interactive execution against production through the CLI and requires explicit confirmation, reducing accidental execution risk during developer and AI-assisted database inspection.

OPEN SOURCE CONTRIBUTIONS

- **Microsoft Playwright:** Investigated and submitted root-cause analysis with reproducible regression test cases for custom reporter AggregateError serialization defects (resolved upstream in v1.61 milestone, [Issue #40856](#)); authored browser runtime workaround for CapsLock keyboard modifier state handling under CDP/BiDi ([Issue #16214](#)).
- **Nightwatch.js:** Isolated and submitted issue reports with reproducible test cases for global hook execution-order lifecycle defects and element locate strategy failures ([Issue #3847](#), [Issue #3530](#)).
- **UnJS (unpdf):** Resolved dual-package type resolution defects in TypeScript dual ESM/CJS declaration exports ([PR #56](#)).

EDUCATION

M.S. in Machine Learning & AI — Liverpool John Moores University	2021 – 2022
Postgraduate Diploma in Machine Learning & AI — IIIT Bangalore	2020 – 2021
Master of Computer Applications — Amity University, Noida	2016 – 2019
Bachelor of Science — Ewing Christian College, Prayagraj	2016